

Abdelmageed Ahmed

Senior Machine Learning Engineer • AI / Data Science

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[Portfolio](#) • [LinkedIn](#) • [Medium](#) • [GitHub](#)

Date of Birth: 01 Mar 1995 • Military Service: Completed

EDUCATION

Master's Degree, Ottawa University - Ottawa, Canada

2022

- Faculty of Engineering - Electrical and Computer Engineering
- **Major:** AI / Data Science
- **Related Coursework:** Machine Learning, NLP, Big Data & Cloud Computing, Deep Learning, Computer Vision, AI for Cybersecurity, and Smart Cities Technologies

Bachelor's Degree, Minya University - El Minya, Egypt

2019

- Faculty of Engineering - Biomedical Engineering
- **Related Coursework:** Biosignals, Image Processing, Digital Electronics, Medical Instruments, and Deep Learning

WORK EXPERIENCE

Senior Machine Learning Engineer, LogicFarm (USA)

Jul 2024 – Present

AI / ML Engineer, KnowledgeNet (Egypt)

Jan 2024 – Jul 2024

AI Engineer, GAIA (UAE)

May 2022 – Jan 2024

Author - "LLM Engineering: Understanding & Deploying Transformer Models" Course ([Udemy](#)) Apr 2023 – Nov 2023

ML Engineer Intern, Microsoft Egypt

Jan 2022 – May 2022

Machine Learning Engineer, Upwork (Freelancer)

Mar 2021 – May 2022

PROJECTS

FinSight - AI-Powered Multi-Agent Financial Research Platform (*side project - [link](#)*)

Ingests SEC filings, live market data, and news for any stock ticker and returns a structured, citation-backed risk report with SWOT, bull/bear analysis, and head-to-head peer comparison.

- Production multi-agent financial analyst that builds the citation-backed risk report from SEC filings, market data, and news - *LangGraph + ChromaDB + OpenAI embeddings + Claude Haiku 4.5.*
- Async backend and typed frontend delivering analyses, chat, and side-by-side comparisons end-to-end - *FastAPI + SQLAlchemy + Alembic on PostgreSQL/Redis, with a Next.js 16 / React 19 / TypeScript / Tailwind / TanStack Query UI. (Frontend is written by Vibe Coding with claude code)*
- Zero-downtime cloud deployment provisioned entirely as code - *AWS (VPC, ECS Fargate, ALB, RDS, ElastiCache, ECR, IAM, Secrets Manager, CloudWatch) via Terraform, shipped through Docker + GitHub Actions over OIDC (no keys).*

AI Dataplate Extraction - Linq Asset Platform (*LogicFarm*)

An AI feature that reads equipment dataplate images, extracts structured field data (manufacturer, model, voltage, etc.) into validated schemas, and learns from user feedback to suppress inaccurate suggestions over time.

- Parallel multi-provider extraction pipeline turning dataplate images into schema-validated fields, running queries concurrently with operator-tunable workers and per-query isolation so one failure never sinks the batch - *LangChain + Anthropic Claude (primary) + OpenAI (fallback), JSON-Schema structured output via agent tool-calling on ThreadPoolExecutor / as_completed via FastAPI.*

AI Insights *(LogicFarm)*

An AI-powered asset intelligence platform that auto-tags incoming maintenance logs, detects anomalies, and generates structured insight and summary reports for facility operations.

- Real-time tagging pipeline that processes incoming entries for daily asset checks, preventive maintenance, repairs, and shift handovers - scoring each against predefined questions and persisting tags with relevance scores to *MongoDB*.

RigGuard *(GAIA)*

An AI-powered real-time safety monitoring system that detects and tracks PPE violations and unsafe behaviors across critical rig zones and triggers instant alerts to prevent workplace incidents.

- Production object detection and tracking pipeline that monitors live rig footage, identifies safety violations in restricted zones, and fires automated alarms upon detection - *YOLO + DeepSORT + Flask + OpenCV + OOP.*

Ragent - Production-Ready Multi-Agent Conversational AI

Routes between a tool-augmented chatbot and a document-grounded RAG agent, with persistent memory, content-safety guardrails, and real-time chat - delivering citation-aware answers over user-uploaded knowledge bases.

- Built a multi-agent orchestration layer that dynamically routes each session between a RAG agent (document Q&A) and a tool-using chatbot, wrapped in input/output safety guardrails and a semantic memory system - *LangGraph + ChromaDB + Sentence Transformers (local, zero-cost embeddings) + OpenAI GPT-4.*
- Engineered an async backend with real-time chat and multi-tenant data isolation - JWT / Google OAuth auth, SHA-256 content-hash duplicate detection, multi-file PDF upload with live progress tracking, and per-session vs. unified knowledge-base modes - *FastAPI + WebSockets + MongoDB + Redis.*
- Integrated external agent tooling with human-in-the-loop control - Google Calendar event creation (user-approval gated), web search, Wikipedia, calculator, and datetime tools, plus configurable guardrails for harmful input/output filtering - *LangChain tools + Serper API + Google Calendar API.* [GitHub](#)

Procurement Data Assistant - Multi-Agent Conversational Analytics System

Turns natural-language questions into validated MongoDB aggregation pipelines and returns citation-ready, conversational answers with full drill-down and CSV/JSON export.

- Production multi-agent system that classifies user intent and routes between a data-query agent (NL → MongoDB) and a chat agent, returning natural-language analytics over a structured procurement knowledge base - *LangGraph orchestration + OpenAI GPT function calling for structured query generation.*
- Engineered a two-tier query engine (fast 100-row summary plus complete 10K-row dataset and exact count) delivering sub-second responses while exposing the full result set and generated MongoDB query through a Technical Details modal with CSV/JSON downloads - solving the “show me all results” gap end-to-end.
- Built a dual-memory system for context-aware responses - short-term conversation history in MongoDB plus long-term semantic recall in ChromaDB with duplicate detection - behind safety guardrails (prompt-injection, PII, XSS, and length validation).
- Shipped a real-time WebSocket chat experience with persistent sessions and history browsing on an async FastAPI backend, plus retry/resend recovery on failed queries.
- Established a quantitative quality bar with an MLflow GenAI evaluation harness - LLM-as-judge scoring queries and responses across weighted criteria (syntax, semantics, efficiency, relevance) on a 100-point scale, with all prompts, schema, metrics, and traces logged for run-over-run comparison. [GitHub](#)

Master's Degree Work - The Power of Knowledge: Building a Knowledge Base from Scratch

- Built a tool that extracts facts from unstructured text and generates RDF triplets.
- Extracted information from Wikipedia articles, infoboxes, and tables using Regex, OpenNRE, Coreference Resolution, NER, and web scraping, and handled user questions translated into SPARQL queries to retrieve short answers. [GitHub](#) [Paper](#) (Sponsored by Microsoft Egypt, Ottawa University, and MCIT Egypt)

TECHNICAL SKILLS & TOOLS

- **LLM / Agentic AI:** RAG, LangChain, LangGraph, OpenAI API, AutoGen, Ollama
- **NLP:** Transformers (BERT, GPT, T5), RNN, LSTM, SpaCy, NLTK
- **Data Science:** Pandas, NumPy, Linear Algebra, Calculus, Statistics, Probability
- **Machine / Deep Learning:** Python, scikit-learn, Keras, TensorFlow, PyTorch
- **Computer Vision:** YOLO, OpenCV, scikit-image, CNNs, GANs, Transfer Learning
- **Deployment & CI/CD:** Terraform, Docker, Kubernetes, FastAPI, Flask, MLflow, GitHub Actions, Azure Pipelines
- **Databases:** Vector DBs (ChromaDB, FAISS, Pinecone, Qdrant), SQL Server, PostgreSQL, MongoDB, Redis
- **Cloud:** Microsoft Azure, AWS, Google Cloud (GCP)
- **Data Visualization:** EDA, Tableau, Power BI, Power Query, Excel, Power Pivot
- **Web Scraping:** Scrapy, BeautifulSoup, Requests, Selenium
- **Other:** PEFT (LoRA, QLoRA), AI Guardrails, Git, WebSocket, HashiCorp Vault

TRAINING, COURSES & CERTIFICATES

Full certificate archive: [Google Drive folder](#)

DELL: Data Science & Big Data Analytics Specialty advanced · associate	Jul 2021
DELL: Data Engineering – Data Scientist Workshop Specialty link	Aug 2021
IBM: Predictive Analytics (SPSS Modeler) Training link	Jun 2021
AWS: Machine Learning Specialty Training link	Apr 2021
Dale Carnegie Institute: Management, Presentation, Marketing & Leadership Training	-
DeepLearning.AI: TensorFlow Developer Professional Certificate	Apr 2021
DataCamp: SQL Server for Database Administrators Track	Jan 2021
Coursera: Data Visualization with Tableau Specialization	Jan 2020
DataCamp: Advanced Deep Learning with Keras in Python	Sep 2019
DataCamp: Data Scientist with Python Track	Sep 2019
DataCamp: Machine Learning with Python Track	Aug 2019
IBM: IBM Data Science Professional Certificate link	Jul 2019